

Middle Convolution of ℓ -adic Perverse Sheaves

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1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations. The symbols below denote types and/or categories depending on your philosophical inclination, where $X : T$ denotes that X is a term of type T , or an object of the category T .

Set	Sets
Set^{fin}	Finite sets
Top	Topological spaces
Grp	Groups
Grp^{top}	Topological groups
Grp^{pro}	Profinite groups
Grp^{ab}	Abelian groups
Ring	Rings (associative, unital, not necessarily commutative)
Ring^{top}	Topological rings
Field	Fields
$\text{Field}^{\text{top}}$	Topological fields
Mod_R	Left modules over $R : \text{Ring}$
Vect_k	Vector spaces over $k : \text{Field}$
$\text{Vect}_k^{\text{fin}}$	Finite-dimensional vector spaces over $k : \text{Field}$
$\text{Rep}_\Lambda(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$
$\text{Rep}_\Lambda^{\text{fin}}(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$, whose underlying topological Λ -modules are finitely generated
Sch	Schemes
Sch_S	Schemes over base scheme $S : \text{Sch}$
Var_k	Varieties over $k : \text{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \text{Field}$, i.e. one-dimensional varieties over k
$\text{Curv}_k^{\text{sm}}$	Smooth curves over $k : \text{Field}$
$\text{Man}_{\mathbb{C}}$	Complex manifolds
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\text{Sh}(X, A)$	Sheaves on $X : \text{Site}$ with values in a category A admitting small limits
$\text{Mod}(X, \mathcal{R})$	Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$, where $\mathcal{R} : \text{Sh}(X, \text{Ring})$
$D^b(A)$	Bounded derived objects of an abelian category A
$D^b(X, k)$	Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$, where $X : \text{Site}$ and $k : \text{Field}$
$X \rightarrow Y$	Morphisms from $X : T$ to $Y : T$

1.1.2 Global Notations. Let $X, Y : \text{Site}$, $k : \text{Field}$, $n : \mathbb{Z}$, $V : \text{Vect}_k$, and $f : X \rightarrow Y$. Also let $T : \text{Top}$ with base point $t : T$. The following notations will be used throughout the write-up.

$(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$	The complex analytification functor
$[n] : D^b(X, k) \rightarrow D^b(X, k)$	The cohomological shift functor by n
$\Gamma(X, -) : D^b(X, k) \rightarrow D^b(\text{Vect}_k)$	The derived global sections functor
$\mathcal{H}^n : D^b(X, k) \rightarrow \text{Sh}(X, \text{Vect}_k)$	The n -th cohomology sheaf functor
$H^n(X, -) : D^b(X, k) \rightarrow \text{Vect}_k$	The n -th hypercohomology functor
$\underline{V} : \text{Sh}(X, \text{Vect}_k)$	The constant sheaf associated with V
$f_* : D^b(X, k) \rightarrow D^b(Y, k)$	The derived pushforward along f
$f^* : D^b(Y, k) \rightarrow D^b(X, k)$	The derived pullback along f
$\otimes_k : D^b(X, k) \times D^b(X, k) \rightarrow D^b(X, k)$	The derived tensor product
$\text{Hom} : D^b(X, k)^{\text{op}} \times D^b(X, k) \rightarrow D^b(X, k)$	The derived sheaf Hom
$\text{End} : D^b(X, k) \rightarrow D^b(X, k)$	The derived sheaf End
$\pi_1^{\text{top}}(T, t) : \text{Grp}^{\text{top}}$	The topological fundamental group, equipped with the discrete topology

where the derived functors are assumed to preserve boundedness.

1.2 Motivations

1.2.1. *Monodromy groups* are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Roughly speaking, they describe how locally defined data change when transported along a loop. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications. We will discuss both the geometric and arithmetic sides, but this write-up will primarily focus on the arithmetic side, with the geometric side serving as motivations and intuitions.

1.2.2. Let $w : \mathbb{C}$ be nonzero, and consider the equation

$$z^2 = w$$

Locally near each w , there are two solutions which we label as $z_+ = \sqrt{w}$ and $z_- = -\sqrt{w}$. Suppose we move w around the unit circle with

$$w(t) = e^{2\pi it} \quad t : \mathbb{R}$$

Then we have $z_+(t) = e^{\pi it}$ and $z_-(t) = -e^{\pi it}$. When t goes from 0 to 1, the two solutions do not return to their original self, but swap places. This induces a permutation S_2 on the set of solutions $\{z_+, z_-\}$. In other words, there is a permutation representation of the fundamental group $\pi_1^{\text{top}}(S^1, 1) \cong \mathbb{Z}$

$$\pi_1^{\text{top}}(S^1, 1) \longrightarrow \text{Aut}_{\text{Set}}(\{z_+, z_-\}) \quad n \mapsto (z_+ \ z_-)^{(-1)^{n+1}}$$

We call this permutation representation the *permutation monodromy representation*, and the image of this representation S_2 the *permutation monodromy group*, of the covering map

$$p : \mathbb{C}^\times \rightarrow \mathbb{C}^\times \quad z \mapsto z^2$$

1.2.3. We can upgrade permutation monodromy representations into linear representations.

1.2.4 A Question in Arithmetic Geometry. Here is a

1.3 Structure of the Write-up

1.3.1. In § 2

1.3.2. In § 3

1.3.3. In § 4

1.3.4. In § 5

1.3.5. In § 6

1.4 Acknowledgements

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1.4.2 Typesetting. The typesetting style of this write-up is inspired by [Kat88].

1.4.3 Declaration of AI Use. The use of generative AI in the preparation of this write-up was limited to the following purposes: polishing wording, phrasing, spelling, grammar, and formatting; offering writing advices; answering mathematical questions from the author to help him learn the subject; searching and citing appropriate references; translating reading materials in French into English; retypesetting old references with archaic typesettings for ease of reading; providing emotional support for the author; providing fitness advice for the author to stay healthy and energetic while working on the write-up.

2 Local Systems and Lisse Sheaves

2.1 Local Systems in Topology

2.1.1 Notation. Define the following collection of notations

$X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \text{Set}^{\text{fin}}$	A finite nonempty subset of closed points of X
$U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	The open subvariety $X \setminus S$
$j : U \rightarrow X$	The inclusion map

$m : \mathbb{N}$ The cardinality of S

2.2 Local Systems in ℓ -adic Arithmetic Geometry

3 Perverse Sheaves and the Six Functor Formalism

4 Middle Convolution and Monodromy Groups

5 Mod- ℓ Monodromy of Hyperelliptic Curves

6 Formalization in Lean

References

[Kat88] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, volume 116 of *Annals of Mathematics Studies*. Princeton University Press, 1988.